Trajectory-exact constrained decoding for masked diffusion models, for any tractable constraint

Working notes

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Abstract

Masked diffusion language models fill in a sequence over several steps, revising their predictions as positions are revealed. Constrained decoders that enforce a constraint one step at a time ignore those revisions and sample from the wrong distribution. TWISTER fixes this for constraints given by deterministic finite automata. These notes separate the fix from the automaton: any structure that can compute the probability of a constraint under a product distribution, and sample from that distribution restricted to the constraint, can take the automaton's place: grammar charts, factor graphs and determinants, with ambiguous grammars allowed. We give the general theorem, the practical failure mode that shows up when the method meets a real model, three remedies with guarantees, experiments on a text diffusion model and a synthetic testbed, and the state of an application to mRNA sequence design.

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1 Introduction

1.1 The problem

A masked diffusion language model [19] builds a sequence of length L from a row of mask symbols over T steps. At each step a *denoiser* reads the current partial sequence and predicts a distribution over the vocabulary at every masked position; a *scheduler* picks which positions to reveal; those positions are filled independently from the predictions. We call this the *native sampler* and write p for the distribution it induces. Within a step, the predictions are a product distribution across positions. Across steps they change, because each denoiser call conditions on the positions revealed so far.

Fix a set C of acceptable outputs. We want to sample from $p(x \mid x \in C)$. Rejection sampling is exact but impractical when $p(C)$ is small, and cheaper methods distort the distribution. Token masking forbids tokens that make C unreachable, which ignores how much probability remains inside C after each choice. *Step-exact decoding* [5] restricts the current product distribution to C and samples from the result. It is also biased, as the following example shows.

Example 1.1 (Step-exact decoding is biased)**.** Let $L = 2$, $T = 2$, reveal position 1 then position 2, and let $C = \{(a, a)\}$: the two tokens must agree. Write $p^1(a)$ for the prediction at position 1 from the all-mask state, $p^2(a \mid \varnothing)$ for the prediction at position 2 from that same state, and $p^2(a \mid a)$ for the prediction at position 2 after position 1 was revealed as a. The native conditional is

$$p(x = (a, a) \mid C) \;\propto\; p^1(a)\, p^2(a \mid a).$$

Step-exact decoding draws position 1 from the restricted product, giving a with probability $\propto p^1(a)\, p^2(a \mid \varnothing)$, and the second step is then forced. The two laws agree only when $p^2(a \mid a)/p^2(a \mid \varnothing)$ is constant in a, that is, when revealing position 1 does not change the model's prediction at position 2.

The ratio $p^2(a \mid a)/p^2(a \mid \varnothing)$ compares the constraint's probability under the updated predictions with its probability under the earlier ones, and it is the importance weight that removes the bias. TWISTER [22] applies this weight at every step to a population of particles and proves that the result targets the native conditional exactly when C is given by a deterministic finite automaton.

1.2 Main result

Main claim.

TWISTER's correction does not require the constraint to be an automaton. It requires a structure that (i) computes the constraint's probability under a product distribution, (ii) samples from that product distribution restricted to the constraint, and (iii) tests membership exactly at the end. We call such a structure a *route*. Automata, grammar charts, factor graphs and determinants are all routes, and each covers constraints the others cannot handle in polynomial time. The route may be ambiguous: a grammar that counts a string once per parse tree still gives the correct target.

The method has two parts. A route handles the computation within one step. A *trajectory layer*, twisted sequential Monte Carlo, handles the dependence across steps and uses the route only through its three operations. Theorem 4.2 shows that this combination is exact.

1.3 Further results

Beyond Theorem 4.2, these notes contain three results.

When the correction is unnecessary. If the denoiser's predictions do not change as positions are revealed, the frozen twist equals the ideal twist and step-exact decoding is already trajectory-exact (Proposition 5.2). The size of the correction on a given model therefore depends on how much its predictions change between steps, a quantity we call re-conditioning strength. It can be measured before running the sampler.

Hidden mass under resampling. The particle estimate $\widehat{Z}$ of $p(C)$ is unbiased for every number of particles. When the current predictions greatly understate the probability of satisfying the constraint at an intermediate step, adaptive resampling discards particles that would later have recovered, and part of $\mathbb{E}[\widehat{Z}]$ is carried by events too rare to observe in practice. On a text diffusion model the average of $\widehat{Z}$ fell 7% short of $p(C)$. Section 6 characterises this *hidden mass* and gives three remedies that keep $\widehat{Z}$ unbiased.

Ambiguous routes and resampling. With an ambiguous grammar, sampling without resampling has low error and resampling increases it. Removing the effect exactly would require counting strings rather than derivations, which is #P-hard in general (Section 7).

Each result is tagged [background] (known, cited), [ours] (proved here) or [proposed] (a direction). Section 12 lists them all. Proofs longer than a few lines are in Appendix A.

2 Setup

Tokens and classes. Let V be the vocabulary and $\text{M} \notin V$ the mask. Constraints are stated over a small alphabet Σ of token *classes* through a map $\phi : V \to \Sigma$ (digit, opening bracket, codon). We write $C \subseteq \Sigma^L$ and say a token sequence x satisfies C if $\phi(x) \in C$. For a partial sequence $x \in (V \cup \{\text{M}\})^L$, $\mathcal{M}(x)$ is its set of masked positions.

Native kernel. One native step from x to x' has probability

$$\kappa_{t+1}(x' \mid x) = s_{t+1}(r \mid x) \prod_{i \in r} p_\theta^i(x'_i \mid x), \tag{1}$$

where $r \subseteq \mathcal{M}(x)$ is the revealed set, s_{t+1} the scheduler, and p_θ^i the denoiser's prediction. The native path law is $p(x_{0:T}) = \prod_t \kappa_{t+1}$, starting from $x_0 = \text{M}^L$.

Reveal schedules. The scheduler is part of the process, not an implementation detail, and two choices are in common use. The *fixed* schedule reveals exactly L/T positions per step, uniformly among the masked ones; this is TWISTER's. The *independent* schedule reveals each masked position with probability $1/(T-t)$, which is the MDLM reverse step [19] under the log-linear noise schedule and leaves a random number of reveals per step. Everything below holds for either, since s_{t+1} cancels in (4). But it must be the *same* scheduler in the proposal and in the native law being conditioned. The two laws are genuinely different: on identical models and constraints, $p(C)$ differs between them by 4 to 10 standard errors at $n = 3 \times 10^5$ samples. mRNAutilus uses the independent schedule, so our mRNA runs do too.

Target. The *Doob path law* $p^\star(x_{0:T}) \propto p(x_{0:T})\mathbf{1}_C(x_T)$ has terminal marginal $p(\cdot \mid C)$, and is a Markov chain with kernel $\kappa_{t+1}h_{t+1}/h_t$, where

$$h_t(x) = \Pr_p(X_T \in C \mid X_t = x)$$

is the *Doob function*. It depends on the denoiser's outputs at every future state and cannot be computed at feasible cost; the methods below replace it with computable twists. We write $h_0 = p(C)$.

Frozen weights and the clamped sum. For a partial sequence x the class weights are $w_i(\sigma \mid x) = \sum_{v:\phi(v)=\sigma} p_\theta^i(v \mid x)$ at masked positions, and $w_i(\sigma \mid x) = [\sigma = \phi(x_i)]$ at revealed ones. The *clamped sum* of a partial sequence x' under the predictions made at x is

$$Z(x'; x) = \sum_{y \in C} \prod_{i \in \mathcal{M}(x')} w_i(y_i \mid x) \prod_{i \notin \mathcal{M}(x')} [y_i = \phi(x'_i)]. \tag{2}$$

Then $\eta_t(x) := Z(x; x)$ is the probability that C holds if every masked position is filled independently from the current predictions. TWISTER uses η_t in place of h_t; we call it the *frozen twist*, since it treats the current predictions as final.

Twisted SMC. This material is standard [6]. A *twist* is a sequence $\psi_0, \ldots, \psi_T$ with $\psi_T = \mathbf{1}_C$, defining intermediate targets $\gamma_t = p(x_{0:t})\psi_t(x_t)$ and *potentials*

$$G_{t+1}(x_t, x_{t+1}) = \frac{\kappa_{t+1}(x_{t+1} \mid x_t)\, \psi_{t+1}(x_{t+1})}{M_{t+1}(x_{t+1} \mid x_t)\, \psi_t(x_t)} \tag{3}$$

for a proposal M. The product $\psi_0 \prod_t G_{t+1}$ telescopes to $p\mathbf{1}_C/q$, so the *final* weight does not depend on the twist; the twist changes only the intermediate weights, and hence which particles survive resampling. The ideal twist is h_t. With K particles, resampling fires when the effective sample size $\mathrm{ESS} = (\sum_k W^k)^2 / \sum_k (W^k)^2$ drops below a threshold, and $\widehat{Z} = \prod_{\text{resamplings}} \left(\frac{1}{K}\sum_k W^k\right) \times \left(\frac{1}{K}\sum_k W^k\right)_{\text{final}}$ is unbiased for h_0 at every K.

3 Routes

Definition 3.1 (Route)**.** A *route* for C provides three operations on weights $w \geq 0$:

- $\textsc{Mass}(w) = \sum_y d(y) \prod_i w_{i,y_i}$, for a fixed $d \geq 0$ with $\operatorname{supp} d \supseteq C$;
- $\textsc{Sample}(w)$, an exact draw from $y \propto d(y) \prod_i w_{i,y_i}$;
- $\textsc{Accepts}(y) = \mathbf{1}_C(y)$, an exact membership test.

It is *exact* if $d = \mathbf{1}_C$, *derivation-weighted* if $d \in \mathbb{N}$ with $\operatorname{supp} d = C$, and a *superset route* if $\operatorname{supp} d \supsetneq C$.

The trajectory layer requires that Sample draw from exactly the measure whose total Mass returns. The function $d(y)$ counts *derivations*: accepting paths of an automaton, parse trees of a grammar. A route is unambiguous when $d \equiv 1$ on C.

3.1 The four families

Table 1 lists the four families; Appendix B gives the recursions, the samplers and their references.

route	cost of one pass	sampler	covers
automaton	$O(L\lvert\delta\rvert)$	forward-backward	motifs, counts, regular expressions, bracket depth
grammar chart	$O(L^3\lvert G\rvert)$	top-down from the chart	context-free constraints, RNA pairing
factor graph [12]	$O(L\lvert\Sigma\rvert^{\omega+1})$	elimination then reverse	copy relations, target structures
determinant	$O(n^3)$	Wilson's algorithm	spanning trees, dependency structures

Table 1: The four route families. ω is the induced width of the elimination order.

Which route is cheapest depends on the representation as well as the constraint. The same-order copy relation $y_i = y_{i+k}$ is width-1 as a factor graph and needs $|\Sigma|^k$ states as an in-order automaton. A compiler estimates the cost of each candidate from compile-time statistics and picks the cheapest. When none fits the budget, or the family has #P-hard counting, it refuses the constraint; it does not approximate.

3.2 Ambiguity depends on the route

The same language can have an unambiguous route and an ambiguous one. Unambiguous Dyck-1 is $S \to (S)S \mid \varepsilon$; ambiguous Dyck-1 is $S \to SS \mid (S) \mid ()$, which generates the same strings but gives a string with several adjacent blocks several parse trees. Whether ambiguity matters depends on the semiring in which the route is evaluated.

Proposition 3.2 ([background]). *Let a structure circuit S represent C. Evaluated on leaves $w \geq 0$, the Boolean semiring answers completability and max-times answers MAP, both regardless of ambiguity; sum-product returns $\sum_y d_S(y) \prod_i w_{i,y_i}$, which is the constrained probability* if and only if *$d_S \equiv 1$ on C.*

Proof. A special case of Wang et al. [23], Thm. 4 and Ex. 2: mapping a Boolean circuit into a semiring is tractable when the semiring is idempotent or the circuit is deterministic. The underlying fact is older and more general: the correctness of a sum-product recursion uses only distributivity and the commutation of nested sums, so any commutative semiring gives a valid algorithm. This is the generalized distributive law of Aji and McEliece [1].

Idempotent addition counts the $d_S(y)$ copies of a string once; ordinary addition counts all of them. An ambiguous route therefore samples from the derivation-weighted law $\propto d(y) \prod_i w_{i,y_i}$, which is why Dang and Ermon call their NFA variant a "soft proxy" [5]. One corollary: Viterbi on the length-L chart computes exact constrained MAP for *every* context-free grammar, inherently ambiguous ones included, in $O(L^3|G|)$. Theorem 4.2 shows that trajectory-level sampling is also exact for ambiguous routes, although single-step sampling is not.

3.3 The tractability boundary

Step-exact decoding needs the constrained probability and its marginals. Both are queries on the *constraint polynomial* $P_C(w) = \sum_{y \in C} \prod_i w_{i,y_i}$, and the following theorem characterises when they are tractable.

Theorem 3.3 ([background]). *Fix Σ and a family $\{C_L\}$, with all bounds polynomial in L and $|\Sigma|$. These are equivalent: (a) P_{C_L} has a polynomial-size arithmetic circuit over $\mathbb{R}$, where subtraction is allowed and the circuit need not be decomposable, deterministic or monotone; (b) a real-RAM algorithm computes $P_{C_L}(w)$ for every $w \geq 0$; (c) a real-RAM algorithm computes all constrained marginals whenever $P_{C_L}(w) > 0$. Given a circuit of size s, P_C costs $O(s)$, all $L|\Sigma|$ marginals together cost $O(s)$, and an exact sample costs $O(Ls)$.*

Proof. (b)⇒(a) is Broadrick et al. [3], Prop. 5.4. Their virtual-evidence marginalisation is query (b) for $\mathbf{1}_C$ after one-hot encoding; we checked that their division-elimination and polynomial-identity steps go through for P_C with correctness only on the positive orthant. (a)⇒(b) is evaluation; marginals come from one reverse-mode pass. (c)⇒(b) and the sampler are self-reducibility under clamping [11].

The theorem requires tractability for arbitrary weights $w \geq 0$. Tractable marginals under hard evidence alone do not imply a small circuit [3], and decoding needs arbitrary weights because the weights are the model's predictions. Decomposable circuits, which include automata, charts and bounded-width factor graphs, do not cover every tractable family:

Remark 3.4 (Spanning trees separate the two [background]). Encode a tree on $\{0, \ldots, n\}$ by head pointers and let C be the arborescences rooted at 0. Then P_C is the directed spanning-tree polynomial: polynomial-size via the matrix-tree theorem, but every *monotone* circuit for it has size $2^{\Omega(n)}$ [10]. Smooth decomposable circuits with non-negative parameters are monotone, so no automaton, grammar or bounded-treewidth route is polynomial here, while exact decoding through a determinant is. The determinant route is included for this case.

If a family has no FPRAS, it has no polynomial-time almost-uniform sampler either [11], so the compiler's refusal of such a family reflects a limit of the problem, not of the implementation.

4 The route-generic algorithm

Write $Z_d(x'; x)$ for the clamped sum (2) with $\mathbf{1}_C$ replaced by the route's d, and $\eta_t(x) = Z_d(x; x)$.

Lemma 4.1 (Step-exact proposal [background]). *The kernel* $M_{t+1}(x' \mid x) = s_{t+1}(r \mid x) \prod_{i \in r} p_\theta^i(x'_i \mid x)\, Z_d(x'; x)/Z_d(x; x)$ *is a probability kernel, sampled by drawing* r *from the native scheduler, drawing a complete class sequence* $y \sim \textsc{Sample}(w(x))$ *with revealed positions clamped, drawing tokens within classes at the positions in* r*, and revealing only those.*

Proof. TWISTER [22], Prop. 3.6, stated for DFAs. The proof uses only that the sampler draws from the measure whose total is $Z_d(x; x)$, so it carries over to any route.

Combining Lemma 4.1 with (3), the scheduler and the token probabilities cancel:

$$G_{t+1}(x, x') = \frac{Z_d(x; x)}{\psi_t(x)} \cdot \frac{\psi_{t+1}(x')}{Z_d(x'; x)}. \tag{4}$$

With TWISTER's choice $\psi_t = \eta_t$ this is $G_{t+1} = Z_d(x'; x')/Z_d(x'; x)$, the mass of the new partial sequence under the new predictions divided by its mass under the old ones. In Example 1.1 it equals $p^2(a \mid a)/p^2(a \mid \varnothing)$.

Algorithm 1 Route-generic twisted SMC (one run with K particles)

Require: route ($\textsc{Mass}, \textsc{Sample}, \textsc{Accepts}$); denoiser; T; K; ESS threshold τ; $\epsilon \in [0, 1)$; $\beta \in (0, 1]$

for each k: $x^k \leftarrow \mathrm{M}^L$, $\eta^k \leftarrow \textsc{Mass}(w(x^k))$, $\log W^k \leftarrow \beta \log \eta^k$; $\log \widehat{Z} \leftarrow 0$
for $t = 0, \ldots, T-1$ **do**
  for each particle k **do**
    draw $r \sim s_{t+1}(\cdot \mid x^k)$ and $y \sim \textsc{Sample}(w(x^k))$; draw tokens within classes on r; $x' \leftarrow x^k$ revealed on r
    $z \leftarrow Z_d(x'; x^k)$ ▷ mass of x' under the old predictions
    if $t < T-1$ **then** call the denoiser at x'; $\eta' \leftarrow Z_d(x'; x')$; $\psi' \leftarrow (\eta')^\beta$
    else $\psi' \leftarrow \textsc{Accepts}(x')$ ▷ exact membership at the last step
    $\log W^k \leftarrow \log W^k + (1-\beta) \log \eta^k + \log \psi' - \log z$; $x^k \leftarrow x'$; $\eta^k \leftarrow \eta'$
  if $t < T-1$ and $\mathrm{ESS}(W) < \tau K$ **then**
    $\log \widehat{Z} \leftarrow \log \widehat{Z} + \log \frac{1}{K} \sum_k W^k$; $\tilde{w} \leftarrow W / \sum_k W^k$; $\pi \leftarrow (1-\epsilon)\tilde{w} + \epsilon/K$
    draw ancestors $a_1, \ldots, a_K$ from π; for each k: $x^k \leftarrow x^{a_k}$, $\eta^k \leftarrow \eta^{a_k}$, $W^k \leftarrow \tilde{w}_{a_k}/\pi_{a_k}$
$\log \widehat{Z} \leftarrow \log \widehat{Z} + \log \frac{1}{K} \sum_k W^k$
return weighted particles $\{(x^k, W^k)\}$ and $\widehat{Z}$

Algorithm 1 replaces TWISTER's automaton with a route and adds two options from Section 6: defensive mixing ϵ and twist power β. With an automaton route, $\epsilon = 0$ and $\beta = 1$, it is TWISTER. At $K = 1$ with the weights ignored it is step-exact decoding [5]. Per particle and step it costs one denoiser call, one route sample and two mass evaluations, the second of which is reused at the next step.

Theorem 4.2 (Route-generic trajectory exactness [ours]). *Assume the denoiser gives every token positive probability, the route satisfies* $\operatorname{supp} d \supseteq C$ *with ambiguity allowed, and the twists satisfy* $\psi_t > 0$ *wherever* $h_t > 0$ *with* $\psi_T = \mathbf{1}_C$*. Then:*

(i) $q(x_{0:T})\, W(x_{0:T}) = p(x_{0:T})\, \mathbf{1}_C(x_T)$ *for every path;*

(ii) the weighted particles target the Doob path law and are consistent for $p(\cdot \mid C)$ *as* $K \to \infty$*;*

(iii) $\mathbb{E}[\widehat{Z}] = h_0$ *for every* $K \geq 1$*, every* $\epsilon \in [0, 1)$*, and every resampling rule whose decision depends only on the current particles and weights;*

(iv) the derivation count affects only the variance, since the factor $d(x_T)$ *from the last proposal step cancels in the last potential.*

Proof idea. By Lemma 4.1 the route's clamped sums appear in both the proposal density and the potential and cancel, leaving a product of twists that telescopes to $p\,\psi_T$. Part (iv) explains why ambiguity does not change the target: at the last step x_T is fully revealed, so $Z_d(x_T; x_{T-1}) = d(x_T)$ appears in the denominator of the final potential and cancels the $d(x_T)$ that the proposal introduced. Full proof in Appendix A.

The two kinds of decoding therefore have different requirements. Step-exact decoding needs an exact route (Theorem 3.3). Trajectory-exact decoding needs a route whose sampler matches its mass, a twist that is positive on completable states, and an exact membership test at the last step; any NFA or CFG chart meets these in polynomial time per particle. This is consistent with the #P-hardness of computing h_t for general CFGs, because SMC never computes h_t and its samples are exact only as $K \to \infty$.

Proposition 4.3 (Terminal-twist pitfall [ours])**.** *If an ambiguous route's own value* $Z_d(x_T; x_T) = d(x_T)$ *is used as the terminal twist in place of* $\mathbf{1}_C$*, the particles target* $\propto p(x_{0:T})\, d(x_T)$*, the derivation-weighted law rather than the one asked for.*

Proof. Replace $\psi_T = \mathbf{1}_C$ by $\psi_T = d$ in the telescoping product of Theorem 4.2(i).

At a fully revealed state MASS returns $d(x_T)$, which equals $\mathbf{1}_C(x_T)$ only for an exact route. The last step must therefore call ACCEPTS.

Remark 4.4 (Relation to prior work)**.** For unambiguous routes Theorem 4.2 is TWISTER's argument [22] (Thm. 4.2) applied to a different route; their proof uses the DFA only to compute Z. Dang et al. [4] observe, for NFAs in autoregressive SMC, that the derivation-weighted distribution is still a valid proposal. Theorem 4.2 combines these for masked diffusion and every route of Section 3. We have not found the terminal-twist pitfall or the analysis of Section 6 stated elsewhere.

4.1 Choice of twist

The proof of Theorem 4.2 uses only two facts: the proposal density is computable, and the twists telescope. The twist can therefore be chosen independently of the proposal. The remedies in Section 6 use this.

Proposition 4.5 (Free intermediate twists [background])**.** *Let* M_{t+1} *be any kernel with computable density, positive wherever* $\kappa_{t+1} h_{t+1} > 0$*, and let* $\psi_t \geq 0$ *be positive where* $h_t > 0$ *with* $\psi_T = \mathbf{1}_C$*. Then* $qW = p\mathbf{1}_C$*, and (ii) and (iii) of Theorem 4.2 hold. The same is true for* random *twists* $\psi_t(x, \xi_t)$*, provided the realisation* ξ_t *is reused in the numerator at step* t *and the denominator at step* $t + 1$*.*

Proof. The deterministic case is the standard twisted-SMC construction [6]; [25] uses it for autoregressive language models. For random twists, include ξ_t in the state; each realisation appears once in a numerator and once in a denominator, so the product still telescopes path by path.

4.2 A diagnostic identity

Lemma 4.6 ([ours]). *Under the conditions of Theorem 4.2,* $\mathbb{E}_q[W] = h_0$ *and* $\mathbb{E}_p[\mathbf{1}_C(x_T)/W] = \Pr_q(x_T \in C)$*, the second expectation taken over native paths. For an exact or derivation-weighted route the right-hand side is 1.*

Proof. By Theorem 4.2(i), $\mathbb{E}_p[\mathbf{1}_C/W] = \sum_{\text{paths}} p\,\mathbf{1}_C\,q/(p\,\mathbf{1}_C)$, the q-probability of ending in C.

The first identity can be checked with proposal runs and the second with native samples, and the two checks detect different errors. Native paths whose weight exceeds every observed proposal weight indicate regions the proposal rarely visits. Running Algorithm 1 with resampling off alongside the main run separates proposal-and-twist error from resampling error; the problem described in Section 6 was found this way.

5 When the correction is unnecessary

With $\psi_t = \eta_t$ the potential (4) is $G_{t+1} = Z_d(x'; x')/Z_d(x'; x)$, which differs from 1 only because the denoiser's predictions change between steps. This section treats the case where they do not.

Definition 5.1 (Trajectory mean-field [ours]). A denoiser is *trajectory mean-field* if $p^i_\theta(\cdot \mid x) = p^i_\theta(\cdot \mid \tilde{x})$ for every i and every pair of partial sequences with $\mathcal{M}(x) = \mathcal{M}(\tilde{x})$: the prediction at a masked position depends on which positions are masked, but not on the values revealed at the others.

Proposition 5.2 (Mean-field collapse [ours]). *Let the denoiser be trajectory mean-field, let* $\psi_t = \eta_t$ *for* $t < T$ *and* $\psi_T = \mathbf{1}_C$*, and take an exact route. Then* $\eta_t = h_t$ *everywhere and* $G_{t+1} = 1$ *almost surely, so with* $K = 1$ *and no resampling,* $\widehat{Z} = \eta_0 = h_0$ *deterministically and the single trajectory is an exact draw from* $p(\cdot \mid C)$*. Step-exact decoding is already trajectory-exact.*

Proof. Write $w_i(\cdot)$ for the common prediction at masked position i, which by Definition 5.1 does not depend on the revealed values. The remaining positions are then filled independently from w whatever order the scheduler chooses, so $h_t(x) = \sum_{y \in C} \prod_{i \in \mathcal{M}(x)} w_i(y_i) \prod_{i \notin \mathcal{M}(x)} [y_i = \phi(x_i)]$, which is (2) at $x' = x$, that is $\eta_t(x)$. For the potential, $Z(x'; x)$ and $Z(x'; x')$ clamp the same positions and differ only in which denoiser call supplied the weights at the masked ones; those weights coincide, so the sums are equal and $G_{t+1} = 1$. At the last step $Z(x_T; x_{T-1}) = \mathbf{1}_C(x_T) = \psi_T(x_T)$ and the ratio is again 1. Constant potentials make $\widehat{Z}$ deterministic, and M is then the Doob kernel. □

For a mean-field model one particle suffices. The constraint still couples the positions, so $p(\cdot \mid C)$ is not a product distribution, but the route's sampler accounts for that coupling exactly in a single draw. The particles and weights correct only for re-conditioning, the change in the denoiser's predictions as positions are revealed. We define the *re-conditioning strength* as the typical size of

$$\log G_{t+1} = \log Z(x'; x') - \log Z(x'; x),$$

the change in the route's mass caused only by the model revising its predictions. It is zero for a mean-field model.

On a model with small re-conditioning strength, step-exact decoding is already close to unbiased and the particle correction changes little. Re-conditioning strength can be measured before building a sampler, by revealing context and recording how the predicted distribution at a held-out masked position changes (Section 8.5).

6 Hidden mass under resampling

Theorem 4.2(iii) gives $\mathbb{E}[\widehat{Z}] = h_0$. If much of this expectation comes from rare events, the average of $\widehat{Z}$ over any feasible number of runs falls short of h_0. On OWT-130M with TWISTER's experimental setup, plain resampling gave $\widehat{Z}/h_0 = 0.930$ (CI 0.910–0.949) over 10^5 runs, while the same configuration without resampling gave 0.998.

6.1 Continuation values and hidden mass

Lemma 6.1 (Continuation value [ours])**.** *Without further resampling, the expected product of the remaining potentials from state x at step s is $V_s(x) = \mathbb{E}_q[\prod_{t\geq s} G_{t+1} \mid x_s = x] = h_s(x)/\psi_s(x)$.*

Proof. From step s on, the proposal kernels and potentials telescope to $p\,\psi_T/\psi_s$ along every path; sum over continuations.

Suppose resampling fires at step s. Writing $W_s^k = \rho_s^k\,\psi_s(x_s^k)$ with ρ_s^k the path ratio p/q up to s, Lemma 6.1 gives

$$\tilde{w}_k \propto \rho_s^k\,\psi_s(x_s^k), \qquad c_k \propto \rho_s^k\,h_s(x_s^k), \tag{5}$$

where c_k is particle k's share of $\mathbb{E}[\widehat{Z} \mid \mathcal{F}_s]$. Resampling allocates offspring in proportion to ψ_s, while each particle's share of the estimate is proportional to h_s. Where the frozen twist understates h_s, resampling is likely to remove particles that carry a large share of the estimate.

Proposition 6.2 (Hidden mass under multinomial resampling [ours])**.** *Under multinomial resampling, particle k has at least one offspring with probability $1-(1-\tilde{w}_k)^K \leq K\tilde{w}_k$, and when it does its contribution is multiplied by about $1/(K\tilde{w}_k)$. Define the hidden mass $H_\varepsilon = h_0^{-1}\,\mathbb{E}[\sum_k c_k \mathbf{1}\{\tilde{w}_k < \varepsilon\}]$. Over R runs, the expected number of runs in which any particle with $\tilde{w}_k < \varepsilon$ leaves an offspring is at most $RK\varepsilon\,\mathbb{E}[N_\varepsilon]$, where N_ε is the number of such particles. When this is much less than 1, the average of $\widehat{Z}$ over the R runs misses about a fraction H_ε of h_0 with high probability, even though $\mathbb{E}[\widehat{Z}] = h_0$.*

Proof. Offspring counts are binomial$(K, \tilde{w}_k)$, giving the first bound; a union bound over particles and runs gives the second. On the event that particle k has n_k offspring they carry weight $\bar{W}$ each, against an expected share c_k that Lemma 6.1 assigns to weight $K\bar{W}\tilde{w}_k$; hence the factor $n_k/(K\tilde{w}_k)$.

Two consequences follow. First, tail diagnostics fitted to observed values of $\widehat{Z}$, such as the generalised-Pareto shape estimate, cannot detect the missing mass, because the events that carry it do not appear in the sample. Second, if $\psi_s = h_s$ then $c_k = \tilde{w}_k \sum_j c_j$ by (5), so $H_\varepsilon \leq K\varepsilon$. Large hidden mass therefore requires twist error of one kind: states at which the frozen twist is much smaller than h_s, that is, over-confident that the constraint has become unreachable.

On OWT-130M, at the first step where resampling would fire, 13% of particles had normalised weight below 10^{-6} and together carried 4.7% of h_0. Their median log twist was -40, and 19% of them had a twist value above 10^{-3} by the end of the run.

Remark 6.3 (Estimating H_ε)**.** Runs with resampling on and off are identically distributed up to the first step resampling would fire. So H_ε is estimable from resampling-off runs: record every particle's weight at every step, find the first step with ESS $< \tau K$, and sum the final contributions of particles whose normalised weight there was below ε.

Remark 6.4 (Synthetic testbeds)**.** η_t evaluates the constraint under the current predictions, and nothing forces those to be consistent with what the denoiser will predict later. A real model can assign a completion almost zero probability at one step and substantial probability a few reveals on. A synthetic denoiser whose predictions depend weakly on context has little hidden mass by construction, so tests on such a model are unlikely to show the problem.

6.2 Three remedies

All three keep $\mathbb{E}[\widehat{Z}] = h_0$ and can be combined. Defensive resampling changes the ancestor distribution, tempering changes the power of the twist, and calibration changes the law under which the twist is computed.

Defensive resampling. Resample ancestors from $\pi = (1-\epsilon)\tilde{w} + \epsilon/K$ instead of $\tilde{w}$, giving each offspring weight $\tilde{w}_a/\pi_a$.

Proposition 6.5 ([ours])**.** *With this rule: $\mathbb{E}[\widehat{Z}] = h_0$; every post-resampling weight is at most $1/(1-\epsilon)$ and every particle survives with probability at least $1-e^{-\epsilon}$* however small its weight*; and for any non-negative continuation values the resampling step adds a conditional relative variance of at most $1/\epsilon$. Under multinomial resampling the corresponding quantity has no bound when a particle with small $\tilde{w}_k$ has a large continuation value.*

Proof. In Appendix A.

The variance bound $1/\epsilon$ and the survival probability $1-e^{-\epsilon}$ hold however far the twist understates h_s. The construction is defensive importance sampling [9] applied to the ancestor distribution. When the twist is accurate, the mixture makes resampling somewhat less efficient. On OWT-130M, $\epsilon = 0.25$ raised $\widehat{Z}/h_0$ from 0.930 to 0.998.

Tempered twists. Raise the frozen twist to a power $\beta \in (0, 1]$.

Proposition 6.6 ([ours])**.** *For $\beta \in (0,1]$ let $\psi_t = \eta_t^\beta$ for $t < T$ and $\psi_T = \mathbf{1}_C$, with the step-exact proposal. Then the potential is $G_{t+1} = \eta_t(x)^{1-\beta}\eta_{t+1}(x')^\beta / Z_d(x'; x)$, parts (ii) and (iii) of Theorem 4.2 hold, and in (5)*

$$\frac{c_k}{\tilde{w}_k} \;\propto\; h_s(x_s^k)^{1-\beta}\Big(\frac{h_s(x_s^k)}{\eta_s(x_s^k)}\Big)^\beta.$$

Proof. In Appendix A.

An over-confidence factor $\delta^{-1} = h_s/\eta_s$ therefore enters the resampling mismatch as $\delta^{-\beta}$. Smaller β uses less of the twist's information: as $\beta \to 0$ particles are resampled on the path ratio alone, and $\beta = 1$ is TWISTER.

Remark 6.7 (When tempering is not enough)**.** Tempering removes hidden mass only if the tempered mismatch is large enough to be sampled. By Proposition 6.2 a particle suppressed by δ^β leaves an offspring in one of R runs only if $RK\delta^\beta \gtrsim 1$, requiring $\beta \lesssim \log(RK)/\log(1/\delta)$. When the twist is over-confident by many orders of magnitude this forces β near 0, which discards the twist's information. Defensive resampling has no such condition, so we use it by default and add tempering when the over-confidence is moderate.

Calibrated twists. Calibration changes the law under which the twist is computed. The frozen twist fills masked positions from the current class weights w_i; any other product law u_i over the masked positions gives a twist that the route evaluates exactly, at the same cost. We use $u_i \propto w_i^\tau$ with $\tau < 1$, which flattens the weights.

Proposition 6.8 (Calibrated twist [ours])**.** *Let $\psi_t = \eta_t^u$ for $t < T$ and $\psi_T = \mathbf{1}_C$, and let $Z_d^{u,r}(x; x)$ be the clamped sum using w on the reveal set r and u elsewhere. Then $M_{t+1}^u(x' \mid x) = s_{t+1}(r \mid x)\prod_{i\in r} p_\theta^i(x_i' \mid x)\, Z_d^u(x'; x)/Z_d^{u,r}(x; x)$ is a probability kernel; it is the step-exact proposal for $\psi = \eta^u$; the potential is $G_{t+1} = \big(Z_d^u(x'; x')/Z_d^u(x; x)\big)\big(Z_d^{u,r}(x; x)/Z_d^u(x'; x)\big)$; and (ii), (iii) of Theorem 4.2 hold. At $u = w$ everything reduces to Lemma 4.1.*

Proof. In Appendix A.

Lemma 6.9 (Calibration floor [ours]). *For $u_i \propto w_i^\tau$ with $\tau \in (0,1]$ and $m = |\mathcal{M}(x)|$, $|\Sigma|^{-m(1-\tau)}\eta_t(x)^\tau \le \eta_t^u(x)$. Hence if the frozen twist is over-confident by a factor δ, the calibrated twist satisfies $h_t/\eta_t^u \le |\Sigma|^{m(1-\tau)}\delta^{-\tau}$.*

Proof. In Appendix A.

Remark 6.10 (Calibration and tempering compared). Tempering and calibration both bound over-confidence by a power of δ, but act differently. Tempering compresses the total η_t and leaves the proposal unchanged. Calibration compresses each completion's weight inside the route sum, so the proposal also looks ahead with u and steers revealed classes towards completions that are plausible under the flattened law. Errors in the two directions have different costs: by (5) a twist that overstates h_s wastes some allocation, whereas one that understates it by orders of magnitude creates hidden mass. This is the importance-sampling rule that proposals should have heavier tails than the target, applied to the intermediate targets. Nothing requires $\tau \le 1$: for a denoiser whose predictions are too flat, $\tau > 1$ is the corresponding correction.

Remark 6.11 (Look-ahead twists). Proposition 4.5 also permits random twists, for instance $\psi_t = \eta_t \hat{G}_t$ with $\hat{G}_t$ the potential product along a speculative rollout, which is an unbiased estimate of h_t when rolled to step T. An unbiased twist does not prevent hidden mass. By (5) resampling misallocates in proportion to h_s/ψ_s, so what matters is the spread of $\log(h_s/\psi_s)$ across the particles of one run, and rollout noise adds to that spread. Moreover $\mathbb{E}\log\hat{G}_t < \log\mathbb{E}\hat{G}_t$, so when the potential product is heavy-tailed a single rollout typically falls well below its mean and understates h_t, which is the direction that creates hidden mass. A biased twist that varies smoothly across particles can therefore hide less mass than an unbiased noisy one. Section 8.3 reports the measurement; we do not use look-ahead twists.

Remark 6.12 (Choosing the calibration). τ, a per-step τ_t, or a richer product law can be fitted from a short pilot run by minimising the empirical variance of $\log\widehat{Z}$. This is controlled SMC [8], with the family restricted to product laws so that every route stays exact. A product law cannot represent correlations between future positions, such as those a copy constraint creates; mixtures of product laws, at one route sum per component, are the next family that keeps exactness.

7 Ambiguity

By Theorem 4.2 an ambiguous route gives the correct target, but it increases variance in two ways: the proposal favours completions with many derivations, and the twist $\eta_d = \sum_y d(y)q(y)$ overstates how reachable C is, in proportion to the ambiguity of the remaining completions.

Proposition 7.1 (Single-step cost of ambiguity [ours]). *Suppose $1 \le d(y) \le D$ on C. Importance sampling from $q_d \propto d\cdot q\cdot\mathbf{1}_C$ to $q(\cdot \mid C)$ uses weights $\propto 1/d(y)$, with ESS fraction at least $4D/(1+D)^2$, and the twist ratio obeys $1 \le \eta_d/\eta \le D$.*

Proof. Kantorovich's inequality applied to $X = 1/d(y) \in [1/D, 1]$; see Appendix A.

The bound is informative when ambiguity is bounded and vacuous when it is not: for the RNA pairing grammar D grows exponentially in L. Early in decoding the twist only sees the ambiguity of completions consistent with the current partial sequence, which can be far smaller than D; whether a per-step measure of that kind controls the ESS of whole trajectories is open (Section 10).

In the experiments of Section 8.1, resampling increased the error on both ambiguous grammars on every resampling arm, and calibration did not remove the increase. The chart's twist is derivation-weighted, so resampling favours partial sequences whose completions have many parse trees over partial sequences with a high probability of completion. A twist that counts strings rather than derivations would remove this effect, but exact string counting is #P-hard for ambiguous grammars in general, so no polynomial-time exact fix is available. We therefore run ambiguous routes without resampling. Fixed target structures compile to factor graphs, which count each string once, and are unaffected.

For CFG constraints on hidden Markov models, the exact constrained likelihood is NP-hard already at ambiguity 2, and an FPRAS exists when ambiguity is polynomially bounded [14]. This suggests approximate string-counting twists under bounded ambiguity, which Proposition 4.5 permits provided the proposal density is exact and the terminal twist is $\mathbf{1}_C$.

8 Experiments

Result files and their artifact identifiers are listed in `docs/status.md` in the repository, and protocol details are in Appendix C. We use three testbeds.

Synthetic. A denoiser over $|V| = 40$ symbols whose context sensitivity is set by a scale parameter, so the exact conditional can be enumerated and the sampler checked against it.

OWT-130M. The released MDLM text model of [19], the setting of TWISTER's own experiments.

mRNAutilus. A 150M-parameter masked discrete diffusion model over full-length mRNA, used here only as the subject of a model diagnostic; the design experiments are future work (Section 9).

8.1 Six constraint families on a real model

The families exercise different routes: COPY (factor graph), REGULAR and COUNTING (automata), DYCK1-UNAMB and DYCK1-AMB (charts, one ambiguous), and RNA-PAIR (the Nussinov–Jacobson pairing grammar [15], ambiguous). OWT-130M, $L = 32$, $T = 16$, $K = 16$. The metric is pairwise total variation against a rejection reference in excess of a matched sampling floor, so 0 is as good as the reference; bootstrap standard errors on the excess are 0.003–0.008.

	step-exact	no resampling		defensive $\epsilon = 0.25$		plain
family	$K = 1$	$\tau = 1$	$\tau = 0.5$	$\tau = 1$	$\tau = 0.5$	$\tau = 1$
copy	0.165	0.136	0.068	0.148	0.040	0.135
regular	0.129	0.026	0.046	0.061	0.057	0.089
counting	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0
dyck1-unamb	0.160	0.133	n/a	0.132	n/a	0.132
dyck1-amb	0.154	0.086	0.091	0.143	0.158	0.151
rna-pair	0.030	0.005	0.006	0.172	0.140	0.212

Step-exact decoding has excess error on five of the six families; by Proposition 5.2 this bias comes from re-conditioning. Particles without resampling reduce it on all five, by between 17% (DYCK1-UNAMB) and 83% (RNA-PAIR). On COPY the largest reduction, to 0.040, needs both calibration and defensive resampling. Resampling increases the error on both ambiguous grammars on every resampling arm, the effect discussed in Section 7. On COPY the Pareto shape $\hat{k}$ of the weights falls from 1.07 (step-exact) to 0.37 ($K = 16$) to 0.08 with calibration.

8.2 Long horizons

The same families at $L = 64$, $T = 64$, $K = 8$, which quadruples the number of weight increments per trajectory. COPY behaves as at $L = 32$ (TV excess 0.095 without resampling, 0.038 with defensive resampling and $\tau = 0.5$), but the arm with the lowest TV excess has $\widehat{Z}/h_0 = 0.902$ (CI 0.827–0.983, $\hat{k} = 0.67$): its normalising-constant estimate falls short of h_0 over the runs performed. COUNTING is exact on every arm. On RNA-PAIR, defensive resampling gives TV excess 0.043 but $\widehat{Z}/h_0 = 0.576$ (CI 0.359–0.858, $\hat{k} = 1.00$, an infinite-mean weight tail).

8.3 Calibrated and look-ahead twists

Synthetic testbed, context scale 8, $K = 4$, 2×10^4 runs per arm, sweeping τ. Efficiency is $1/(\text{relative s.e.}^2 \times \text{denoiser evaluations per run})$, normalised to the resampling-off arm at $\tau = 1$; values below 1 indicate a resampling penalty.

τ	no resampling	plain	defensive $\epsilon = 0.25$	hidden mass $< 10^{-2}$	resamplings/run
1	1.00	0.14	0.47	7.8%	1.78
0.7	0.99	0.32	0.78	3.2%	1.69
0.5	1.02	0.63	1.09	0.9%	1.56
0.35	1.01	1.04	1.15	0.16%	1.40
0.2	1.03	1.09	1.04	0.15%	1.17

All fifteen arms give $\widehat{Z}/h_0$ between 0.979 and 1.008, consistent with Proposition 6.8. With resampling on, efficiency increases as τ decreases to 0.35 (plain 0.14 to 1.04, defensive 0.47 to 1.15), and the hidden mass below 10^{-2} falls from 7.8% to 0.16%, consistent with Lemma 6.9.

Look-ahead twists were tested in the same testbed. Plain resampling with a one-step or a full look-ahead twist reached efficiency 0.001. The fraction of h_0 hidden below 10^{-6} rose from 0.0004 with the frozen twist to 0.069 with full look-ahead. Averaging log-gains over four rollouts reduced it to 0.006, 20 times the frozen twist's 0.0003 in the same run. The rollout noise widens the spread of $\log(h_s/\psi_s)$ by more than it reduces the twist's bias.

Remark 8.1 (Calibration with resampling off)**.** Calibration's effect with resampling off is not explained by smoother per-step weight increments. A $K = 1$ trace decomposition shows summed per-step log-weight variance falling $11.1 \to 6.0$ as τ goes $1 \to 0.5$, while the total path log-weight standard deviation barely moves ($3.278 \to 3.298$) and the relative standard error of $\widehat{Z}$ is flat ($0.0128 \to 0.0132$). Across families at $K = 16$ with resampling off, the ratio of relative standard errors ($\tau = 0.5$ vs $\tau = 1$) is 1.02, 1.17, 0.30 and 0.66 on PREFIX, COPY, COUNTING and FORBIDDEN. The effect depends on the family, is sometimes negative, and its mechanism is open.

8.4 Ambiguity

Synthetic testbed with grammar routes. At $L = 4$ and $L = 6$ the sampled law matches the exact conditional to total variation $\leq 4 \times 10^{-16}$ for both ambiguous and unambiguous grammars, as Theorem 4.2 requires. Ambiguity affects the weights. At $K = 16$ the ESS fraction over $L = 8$ to 32 holds at 0.93–0.97 for unambiguous Dyck-1 but falls from 0.92 to 0.34 for the ambiguous grammar and from 0.89 to 0.30 for RNA-PAIR, tracking the mean log derivation count, which reaches 4.4 and 5.3.

8.5 Is there anything to correct on an mRNA model?

A direct measurement of re-conditioning strength (Section 5) before committing to the application. Holding one codon position masked and revealing the rest of a coding sequence in four stages, the predicted distribution at the held-out position moves by total variation 0.337 end to end, or 0.205 restricted to the synonymous codons of the true residue, with the largest single movement, 0.291, in the first quarter of reveals, which is where step-exact decoding commits. A trajectory mean-field model would give 0 at every stage. Two related measurements: randomising the whole context moves the prediction by 0.336, and a synonymous shuffle preserving the encoded protein still moves it by 0.261. The model's joint is not a product of per-position codon tables.

Its *mode*, however, is. On held-out synonymous-codon prediction ($n = 548$ paired positions) the model is top-1 correct 39.8% of the time against 38.7% for a human codon-frequency lookup table and 29.5% for uniform choice among synonyms; McNemar's test gives $z = +0.53$, not significant. These findings are compatible and say different things: a product model reproduces the marginal mode and cannot reproduce the joint.

Caveats: one coding sequence, which appears in the model's own configuration and may be in its training data; the perturbed contexts are somewhat out of distribution; $n = 6$ held-out positions for the re-conditioning figures. These are order-of-magnitude indications and should be repeated across several real human coding sequences.

9 Application: mRNA sequence design

9.1 The design problem

A protein is fixed, since it is the product, but the genetic code is degenerate, so most amino acids have two to six synonymous codons and a 300-residue protein has on the order of 10^{150} coding sequences. They differ in translation efficiency, stability, folding near the start codon, immunogenicity, and manufacturability: whether a restriction site breaks the cloning step, whether a homopolymer run causes synthesis errors. Choosing well within that equivalence class is codon optimisation.

What makes this a good testbed, beyond the application itself, is that mRNA design has something most generation tasks lack: a deterministic, cheap, non-learned scorer. ViennaRNA computes exact minimum free energy and ensemble defect. Candidates can be evaluated without trusting a learned reward model, which is the usual weak point of controllable-generation evaluations.

9.2 How the system is layered

Three layers, with strict dependence in one direction. *Layer 1* is everything in Sections 3, 4 and 6 and knows nothing about biology. *Layer 2* is the codon alphabet, the genetic code, and constraint builders that return layer-1 route objects over an abstract alphabet. *Layer 3* is the model adapter.

Protein identity enters *through the route*, as per-position allowed-codon masks, never by renormalising the model's predictions. This is the methodological point of the whole application: mRNAutilus's own conditional sampler imposes protein identity by masking logits at each step, which is exactly the step-local projection that Example 1.1 shows to be biased. Our route imposes the same requirement without touching the trajectory law.

Constraints compile as expected. For a 60-residue protein: motifs alone give a 20-state automaton; motifs plus a GC band give 2,174 states in 1.0 s; motifs, GC band and a 7-base-pair hairpin give 968 states, domain 213, induced width 3, in 0.5 s. The automaton-times-structure product initially needed 1792^3 cells and now enumerates only feasible (codon, state) pairs.

9.3 What to measure, and what not to

A survey of the codon-optimisation literature settled a question that changes what we claim. **No published learned codon-optimisation method reports per-position codon reconstruction accuracy against a frequency baseline as a primary result.** The single instance in the literature is an unnumbered density plot at nucleotide resolution in which the frequency baseline is barely separated from uniform random synonymous choice. The field reports distributional agreement with held-out natural sequences, positional rare-codon placement, and downstream property prediction instead. Our flat top-1 result in Section 8.5 is therefore not a failure to reproduce published gains; it is a metric those gains were never claimed on. It stays as a one-line sanity check that the model is not broken, which is what it established.

The constrained families are the headline, because that is where rule-based construction fails outright rather than merely underperforming: target secondary structure, GC content in sliding windows, motif avoidance where the forbidden motif straddles a codon boundary, and conjunctions of these. A frequency table cannot satisfy any of them; a greedy repair heuristic satisfies them while distorting the distribution in ways we can measure exactly.

Planned measurements, in order of how directly they test our claim:

1. *Exactness and coverage.* The fraction of generated sequences satisfying the amino-acid constraint exactly (1 by construction for us, measurably less for step-local projection), and the total variation between our sampler's law over the synonymous lattice and the model's true conditional, on short coding sequences where the lattice can be enumerated. This is the claim no incumbent can state.
2. *Positional profile against three controls.* CAI, length-normalised MFE, GC and GC3, and a positional rare-codon profile distance to the natural sequence, against a frequency-table draw, a max-CAI greedy construction, and the natural sequence. The positional term is the discriminating one: it rewards reproducing *where* rare codons sit, which a product model is structurally incapable of.
3. *Best-of-K under a non-learned scorer*, with diversity at fixed K reported alongside, against mRNAutilus's own mode-seeking sampler.

9.4 Why a diverse library, specifically

Three reasons that stack. Every computational mRNA score correlates imperfectly with what happens in cells, so taking the argmax under an imperfect score bets everything on its top-ranked point, whereas synthesising many diverse high-scoring candidates samples the score's upper region. Optimisers with restarts give near-duplicates around one mode, while a calibrated sampler gives spread at controlled quality. And every candidate must be manufacturable, so a library where 30% of members violate a constraint wastes 30% of the synthesis budget. Hard guarantees plus diversity is the combination existing methods do not offer together: optimisers give quality without diversity, unconstrained samplers give diversity without guarantees, and step-local projection gives guarantees with a silently distorted distribution.

9.5 State of the implementation

The model adapter is written and the encoder validated: the vendored transformer block is bit-identical to upstream's on identical inputs, after fixing a reversed SwiGLU gate order and a missing token-dropout rescaling. An end-to-end smoke test samples constrained coding sequences, with all particles live, every sequence translating to the target peptide, and none containing a forbidden restriction site. The design experiments have not been run.

10 Open directions

10.1 Can the twist be made less over-confident without adding noise?

By Proposition 6.2, over-confidence of the frozen twist causes hidden mass, and by Proposition 4.5 the intermediate twists can be chosen freely. Calibration (Proposition 6.8) reduces hidden mass and look-ahead twists increase it (Section 8.3). It is open whether some twist is both smooth across particles and closer to h_t. Candidates are per-step τ_t fitted from a pilot run, mixtures of product laws for correlated constraints at one route sum per component, and learned twists $\psi_\vartheta \approx h_t$ fitted to native rollouts as in [25], with η_t as a feature.

10.2 Calibration with resampling off

With resampling off, calibration at $\tau = 0.5$ reduces the relative standard error of $\widehat{Z}$ by a factor of 3.3 on COUNTING and increases it by a factor of 1.17 on COPY (Section 8.3). Smoother per-step increments do not explain the difference. Identifying the mechanism might show which constraints the calibrated proposal suits.

10.3 Can hidden mass be bounded online?

Proposition 6.2 expresses hidden mass through h_s/ψ_s, which cannot be observed, and tail diagnostics fitted to observed $\widehat{Z}$ cannot detect it. An estimate or upper bound computable during a run, perhaps from a small population of particles that is never resampled, would replace the separate resampling-off experiment now needed to detect it.

10.4 Does bounded per-step ambiguity control trajectory ESS?

Proposition 7.1 bounds the single-step cost by the global derivation range D, which is vacuous when D grows exponentially. Early in decoding the twist sees only the ambiguity of completions consistent with the current partial sequence, which can be much smaller. It is open whether a local measure of this kind bounds the ESS of whole paths.

10.5 Superset routes

For a context-free C, propose from a regular superset $R \supseteq C$ with twist $\psi_t = Z_R(x; x) \cdot \mathrm{comp}_C(x)$, where comp_C tests completability. By Proposition 4.5 this is exact provided $C \subseteq R$ and comp_C never rejects a completable state. Reporting 1 for a dead state only wastes a particle, while a regular *subset* is biased unless the bound provably covers every string of length L. Pruning converts one accept-or-reject at the end into deaths at individual steps, which resampling can repair, provided R is tight enough to keep the per-step death rate well below $1 - 1/K$. A [proposed] variant: rather than letting a dead proposal kill its particle, redraw until comp_C accepts, and use an unbiased estimate of the single-draw acceptance probability in the potential, which is valid by the random-twist part of Proposition 4.5.

10.6 Where does exact sampling overtake mode-seeking in best-of-K?

Exact conditional sampling should eventually beat mode-seeking decoding as K grows, since the mode is a single point and the sampler explores the upper region of the score. Where the crossover lies, and how it depends on the scorer's correlation with the true objective, is open, and is what the mRNA best-of-K experiment (Section 9) is designed to measure.

10.7 Beyond a fixed protein

The route machinery is indifferent to biology. The natural next targets are better cases for the method than coding-sequence design: 5′UTR design has no protein constraint at all, so the model's joint carries all the information, and translation initiation depends on structure near the cap and start codon: global constraints that rules handle badly. Constraints spanning the UTR–CDS boundary are where codon-table methods are structurally incapable rather than merely worse. Structured text generation is the same algorithm with a different compile target.

11 Related work, by component

The trajectory correction. TWISTER [22] is the direct predecessor: twisted SMC for masked diffusion with DFA constraints, including the step-exact proposal (Lemma 4.1) and the exactness theorem for that case. Zhao et al. [25] apply twisted SMC to autoregressive language models with learned twists. Dang et al. [4] observe that a derivation-weighted NFA proposal remains valid in autoregressive SMC. The underlying Feynman–Kac theory, including unbiasedness of $\widehat{Z}$ under adaptive resampling, is [6].

Step-local constrained decoding. Dang and Ermon [5] give the automaton-based step-exact decoder that Example 1.1 shows to be biased across steps; it is our $K = 1$ arm. Su and Zhang [21] treat constraints as graphs rather than chains, which corresponds to the factor-graph route here.

What is tractable in one step. The circuit characterisation is Broadrick et al. [3]; the semiring behaviour under ambiguity is a special case of the compositional atlas of Wang et al. [23]. Jerrum, Valiant and Vazirani [11] give the sampling-to-counting reduction the compiler uses to decide refusals, and Jerrum and Snir [10] the monotone lower bound that motivates the determinant route. Marzouk and de la Higuera [14] give hardness results for CFG-constrained inference in hidden Markov models and an FPRAS under bounded ambiguity.

The individual routes. Grammar charts and top-down sampling are classical; the RNA pairing grammar is Nussinov and Jacobson [15]. The determinant route uses the matrix-tree marginals of Koo et al. [13] and Wilson's sampler [24].

The remedies. Defensive resampling adapts defensive importance sampling [9] to the ancestor distribution. Calibrated twists are a restricted, closed-form member of the controlled-SMC family [8], which fits twists by regression within a parametric family; restricting to product laws keeps every route exact.

The model. Masked diffusion language models follow Sahoo et al. [19], whose reverse step defines the independent schedule of Section 2.

Who else composes constraints of this kind. Two axes separate the prior work: which constraint classes a method can express, and which semiring it evaluates them in. On expressiveness, no decoder matches the set in Table 1: TWISTER [22] is trajectory-exact but automaton-only; Su and Zhang [21] argue for graphs over chains and give the factor-graph case; grammar-constrained decoding [7] handles context-free constraints for autoregressive models by per-step masking, which

is biased in the same way Example 1.1 describes, and grammar-aligned decoding [18] removes that bias by iteratively reweighting across samples rather than within a trajectory.

The closest prior construction is not in the decoding literature at all. LinearDesign [20] encodes the synonymous-codon design space of a protein as a weighted DFA and runs *lattice parsing* against an RNA folding grammar: the automaton-times-grammar conjunction of Appendix B, built for exactly our application. The difference is the semiring. LinearDesign evaluates the intersection in max-times and returns the single optimum of a hand-written objective (free energy traded against codon adaptation); we evaluate it in sum-product and return samples from a learned model's conditional. By Proposition 3.2 that difference is not cosmetic: the max semiring is correct whatever the grammar's ambiguity, while the sum semiring counts each string once per derivation. LinearDesign can intersect with an ambiguous folding grammar and ignore the issue; we cannot, which is what Section 7 is about. So the honest statement of novelty is narrow and worth stating precisely: the conjunction is not new, and performing it in the sum semiring over a learned distribution is.

12 Summary

Summary.

The trajectory-level correction for masked diffusion decoding is independent of the constraint representation: any route with a matching mass and sampler, plus an exact terminal membership test, gives exact conditional sampling (Theorem 4.2), ambiguous grammars included. If the denoiser's predictions do not change as positions are revealed, step-exact decoding is already exact (Proposition 5.2); the size of this change, the re-conditioning strength, can be measured before building a sampler. Resampling can leave part of $\mathbb{E}[\widehat{Z}]$ in hidden mass (Proposition 6.2); defensive resampling bounds the added variance independently of the twist, and calibration reduces hidden mass by an amount that depends on the constraint family. On ambiguous routes resampling increased the error in our experiments, and the string-counting twist that would avoid this is #P-hard to compute in general.

result	status	source or note
Semirings and ambiguity (Proposition 3.2); MAP for any CFG	background	[23]
Circuit characterisation (Theorem 3.3)	background	[3, 11]
Spanning trees separate monotone from general circuits (Remark 3.4)	background	[10]
Step-exact proposal (Lemma 4.1); free twists (Proposition 4.5)	background	[22, 6]
Routes with single-pass exact samplers; compiler with refusal	ours (system)	Section 3
Route-generic trajectory exactness (Theorem 4.2)	ours	extends [22]; compare [4]
Terminal-twist pitfall (Proposition 4.3)	ours	
Mean-field collapse (Proposition 5.2); re-conditioning strength	ours	case where step-exact is already exact
Two-sided identity (Lemma 4.6)	ours	diagnostic
Continuation value and hidden mass (Lemma 6.1 and Proposition 6.2)	ours	measured at 7% of h_0
Defensive resampling (Proposition 6.5)	ours	compare [9]
Tempered intermediate twists (Proposition 6.6)	ours	
Calibrated twists and their floor (Proposition 6.8 and Lemma 6.9)	ours	compare [8]
Single-step cost of ambiguity (Proposition 7.1); no resampling rule	ours	exact fix is #P-hard
Look-ahead twists measured and dropped	ours	Section 8.3
Superset routes with Boolean pruning	ours	Section 10
Online hidden-mass guard; trajectory ESS under ambiguity	proposed	Section 10

A Proofs

Proof of Theorem 4.2. (i) By Lemma 4.1, $q = \prod_t \kappa_{t+1} Z_d(x_{t+1}; x_t)/Z_d(x_t; x_t)$. Multiplying by (4), the Z_d factors cancel and the twists telescope to $qW = \psi_0 \prod_t \kappa_{t+1}\psi_{t+1}/\psi_t = p\,\psi_T = p\,\mathbf{1}_C$. The divisions are legitimate on the relevant paths: if x' can still be completed to an element of C then some $y \in C$ agrees with x', so $d(y) > 0$ and hence $Z_d(x'; x) > 0$ under the positivity assumption, giving $M_{t+1}(x' \mid x) > 0$. If $\psi_t(x) = 0$ the particle already has weight zero and we set $0/0 = 0$.

(ii) follows from (i) by the standard Feynman–Kac argument [6].

(iii) Unbiasedness under multinomial resampling with adaptive decisions is standard [6]. For $\epsilon > 0$, let $\mathcal{F}$ be the particle system just before a resampling step, with weights W^k and mean $\bar{W}$. For any f,

$$\mathbb{E}\Big[\bar{W}\,\frac{1}{K}\sum_j \frac{\tilde{w}_{a_j}}{\pi_{a_j}} f(x^{a_j}) \;\Big|\; \mathcal{F}\Big] = \bar{W}\sum_k \tilde{w}_k f(x^k) = \frac{1}{K}\sum_k W^k f(x^k),$$

because $\pi_k > 0$ wherever $\tilde{w}_k > 0$. A resampling step therefore leaves the conditional expectation of the weighted particle measure unchanged, whether or not it fires. Iterating gives $\mathbb{E}[\widehat{Z}] = \mathbb{E}_q[W] = h_0$ by (i).

(iv) At the last step x_T is fully revealed, so $Z_d(x_T; x_{T-1}) = d(x_T)$, and (4) gives $G_T =$

$Z_d(x_{T-1}; x_{T-1})\mathbf{1}_C(x_T)/\big(\psi_{T-1}(x_{T-1})\, d(x_T)\big)$: the derivation count enters the denominator and cancels the one the proposal introduced. □

Proof of Proposition 6.5. (i) is Theorem 4.2(iii). (ii) follows from $\pi_k \geq (1-\epsilon)\tilde{w}_k$ and $\pi_k \geq \epsilon/K$. For (iii), let $Y = (\tilde{w}_a/\pi_a)\, g_a$ with $a \sim \pi$. Then $\mathbb{E}Y = \sum_k \tilde{w}_k g_k$ and $\mathbb{E}Y^2 = \sum_k \tilde{w}_k^2 g_k^2/\pi_k \leq (K/\epsilon)\sum_k (\tilde{w}_k g_k)^2$. The variance of an average of K independent copies is at most $\mathbb{E}Y^2/K$, and $\sum_k (\tilde{w}_k g_k)^2 \leq (\sum_k \tilde{w}_k g_k)^2$ because $g \geq 0$. Dividing gives the bound $1/\epsilon$. Under multinomial resampling the same ratio is $\frac{1}{K}\big(\sum_k \tilde{w}_k g_k^2/(\sum_k \tilde{w}_k g_k)^2 - 1\big)$, unbounded when a particle with small $\tilde{w}_k$ has large g_k. Taking $g_k = V_s(x^k)$ from Lemma 6.1 covers the situation of Proposition 6.2. □

Proof of Proposition 6.6. $\eta_t^\beta > 0$ wherever $\eta_t > 0$, which includes every completable state, so Proposition 4.5 applies. The potential is $G_{t+1} = \eta_t(x)^{1-\beta}\eta_{t+1}(x')^\beta / Z_d(x'; x)$, and the expression for $c_k/\tilde{w}_k$ is (5) with $\psi_s = \eta_s^\beta$. □

Proof of Proposition 6.8. (i) Fix r. $Z_d^u(x'; x)$ depends on the tokens revealed on r only through their classes, so summing $\prod_{i\in r} p_\theta^i(x_i' \mid x)\, Z_d^u(x'; x)$ over tokens on r gives $\sum_c \prod_{i\in r} w_i(c_i \mid x)\, Z_d^u(x_c'; x) = Z_d^{u,r}(x; x)$, where x_c' is x with classes c on r. The marginal on r of the measure SAMPLE draws from is $\prod_{i\in r} w_i(c_i \mid x)\, Z_d^u(x_c'; x)/Z_d^{u,r}(x; x)$, and tokens within classes contribute $p_\theta^i(x_i' \mid x)/w_i(c_i \mid x)$. (ii) is the form of M^u. (iii) is (3) with $\kappa/M^u = Z_d^{u,r}(x; x)/Z_d^u(x'; x)$. (iv) $u_i > 0$ wherever $w_i > 0$, so $\eta_t^u > 0$ wherever $\eta_t > 0$, which includes every state with $h_t > 0$, and $M^u > 0$ wherever $\kappa h > 0$; Proposition 4.5 applies. □

Proof of Lemma 6.9. Since $\sum_\rho w_i(\rho) = 1$ and $0 < \tau \leq 1$, Hölder gives $1 \leq \sum_\rho w_i(\rho)^\tau \leq |\Sigma|^{1-\tau}$, so $|\Sigma|^{-(1-\tau)} w_i^\tau \leq u_i \leq w_i^\tau$ at every masked position. Multiply over positions and sum against d. For the lower bound use $(\sum_j b_j)^\tau \leq \sum_j b_j^\tau$, counting each derivation of y as one term. The last claim uses $h_t \leq 1$. □

Proof of Proposition 7.1. Let $X = 1/d(y)$ with $y \sim q_d$, so $X \in [m, M]$ with $m = 1/D$, $M = 1$, and write $\mu = \mathbb{E}X$. From $(X - m)(M - X) \geq 0$ we get $\mathbb{E}X^2 \leq (m + M)\mu - mM$. The quadratic $\frac{(m+M)^2}{4mM}\mu^2 - (m+M)\mu + mM$ has zero discriminant, hence is non-negative, giving $\mathbb{E}X^2 \leq \frac{(m+M)^2}{4mM}\mu^2$ (Kantorovich). Substituting $m = 1/D$, $M = 1$ gives the ESS bound; the twist bound is immediate from $1 \leq d \leq D$. □

B Route implementations

Automata. For (Q, q_0, δ, F) over Σ, the forward recursion $\alpha_0(q) = [q = q_0]$, $\alpha_i(q) = \sum_{q', \sigma: q \in \delta(q', \sigma)} \alpha_{i-1}(q') w_{i,\sigma}$, MASS $= \sum_{q\in F} \alpha_L(q)$ costs $O(L|\delta|)$. SAMPLE is forward filtering, backward sampling: draw $q_L \propto \alpha_L(q)[q \in F]$, then for $i = L, \ldots, 1$ draw (q_{i-1}, σ_i) with probability $\propto \alpha_{i-1}(q_{i-1}) w_{i,\sigma_i} [q_i \in \delta(q_{i-1}, \sigma_i)]$. For a DFA $d = \mathbf{1}_C$; for an NFA $d(y)$ counts accepting paths. Compiles: forbidden and required motifs via Aho–Corasick, counting conditions ($O(Lk)$ states), regular expressions over classes, bounded bracket depth, sliding-window all-different, and products of these. The forward pass must be in float64. In float32 it returned $Z = 0$ for a feasible constraint at log-mass -113.5.

Grammar charts. For a grammar in Chomsky normal form, the inside algorithm

$$\beta(A, i, i{+}1) = \sum_\sigma [A \to \sigma] w_{i,\sigma}, \qquad \beta(A, i, j) = \sum_{A \to BC} \sum_{i<k<j} \beta(B, i, k)\beta(C, k, j), \qquad \text{MASS} = \beta(S, 0, L)$$

costs $O(L^3|G|)$; leaves for all positions come from one matrix product of the denoiser output with the lexical map. SAMPLE pushes $(S, 0, L)$ and repeatedly pops (A, i, j): at a leaf draw $\sigma \propto [A \to \sigma] w_{i,\sigma}$, otherwise draw $(A \to BC, k) \propto \beta(B, i, k)\beta(C, k, j)$ and push both children. This draws each parse tree with probability proportional to its weight, so the sampled string has probability $\propto d(y) \prod_i w_{i,y_i}$. On a GPU all spans of one length are processed together. Per-nonterminal span limits (a subtree rooted at a listed nonterminal covers at most W positions) cut the work to $O(L^2 W|G|)$ when every nonterminal outside a right-branching spine is bounded.

Factor graphs. With 0/1 factors and an elimination order of induced width ω, variable elimination gives MASS in $O(L|\Sigma|^{\omega+1})$, and SAMPLE eliminates in that order and samples in reverse, each bucket table being the exact conditional. Derivations are assignments, so $d = \mathbf{1}_C$, and fixed target structures are unaffected by ambiguity. The motivating case is the same-order copy relation [21], of width 1: MASS $= \prod_{i \le k} \sum_\sigma w_{i,\sigma} w_{i+k,\sigma} \times \prod_{i > 2k} \sum_\sigma w_{i,\sigma}$. Elimination cost is governed by the induced width of the order, and the smallest width over all orders is the treewidth; finding an optimal order is NP-hard [2], so we use the greedy minimum-degree heuristic (George and Liu, SIAM Review 31(1):1–19, 1989).

Determinants. Let $w_m(h)$ be the weight that node m has head h, and $\hat{L}$ the reduced Laplacian with $\hat{L}_{mm} = \sum_{h \ne m} w_m(h)$ and $\hat{L}_{hm} = -w_m(h)$. The directed matrix-tree theorem gives MASS $= \det \hat{L}$, and differentiating $\log\det \hat{L}$ gives the marginals $\mu_m(h) = w_m(h)([\hat{L}^{-1}]_{mm} - [h \ne 0][\hat{L}^{-1}]_{mh})$ [13]; both cost $O(n^3)$. SAMPLE is Wilson's algorithm [24]: for each node not yet in the tree, run the random walk moving $m \to h$ with probability $w_m(h) / \sum_{h'} w_m(h')$ until it meets the tree, erase loops, and add the path.

From classes to tokens. A route draws a class sequence y; each revealed position then draws a token within its class, $v \propto p^i_\theta(v \mid x)[\phi(v) = y_i]$. The combined token-level law is $\prod_i p^i_\theta(v_i \mid x)\mathbf{1}_C(\phi(v))/Z$, which is what the trajectory layer requires. Class masses must be computed by exact per-class log-sum-exp rather than a floored sum: flooring at 10^{-38} inflates rare classes and biases the weights in the rare-event regime.

Conjunctions. Automaton with automaton gives a product automaton. Automaton with grammar gives a grammar, by the Bar-Hillel triple construction, of size at most $|R|Q^3$ and usually far less. Automaton with factor graph gives a factor graph over augmented variables $z_i = (y_i, s_i)$, with the automaton contributing a chain of pairwise factors; restricting to feasible (symbol, state) pairs per position is what makes this practical. Two grammars, or a grammar with a factor graph, are refused, as is any conjunction whose counting problem is #P-hard.

C Experimental protocol

Estimators and diagnostics. $\widehat{Z}$ ratios are reported against h_0 estimated from a large native sample, with bootstrap confidence intervals that include the binomial uncertainty in h_0. $\hat{k}$ is the generalised-Pareto shape of the upper weight tail, fitted as in Pareto-smoothed importance sampling with the Zhang–Stephens estimator; $\hat{k} > 0.5$ indicates infinite variance, > 0.7 that the plain mean is unreliable, > 1 infinite mean. Sample quality is pairwise total variation between empirical joint class distributions of every position pair, averaged over pairs, measured against a rejection reference and reported in excess of a matched sampling floor.

Correctness checks. Each route is verified against brute-force enumeration at small L; the sampler is verified for $\mathbb{E}[\widehat{Z}] = h_0$; golden regression values pin the RNG stream so that refactors cannot silently change results. These checks found four bugs, none of which caused a crash: a float32 automaton forward pass returning $Z = 0$ at log-mass -113.5; class probabilities floored at 10^{-38} biasing weights in the rare-event regime; padded columns counted as reveals under the independent schedule, biasing $\widehat{Z}$ by about 30%; and a scheduler refactor that moved the reveal draw ahead of the route sample, changing the RNG order.

Reproduction. Result files and artifact identifiers are listed in `docs/status.md`. The test suite covers compilation, CPU and batched samplers, SMC, resampling, diagnostics, exact laws, alphabets, and the biology layers.

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